life.augmented

Power Tree Designer

Report

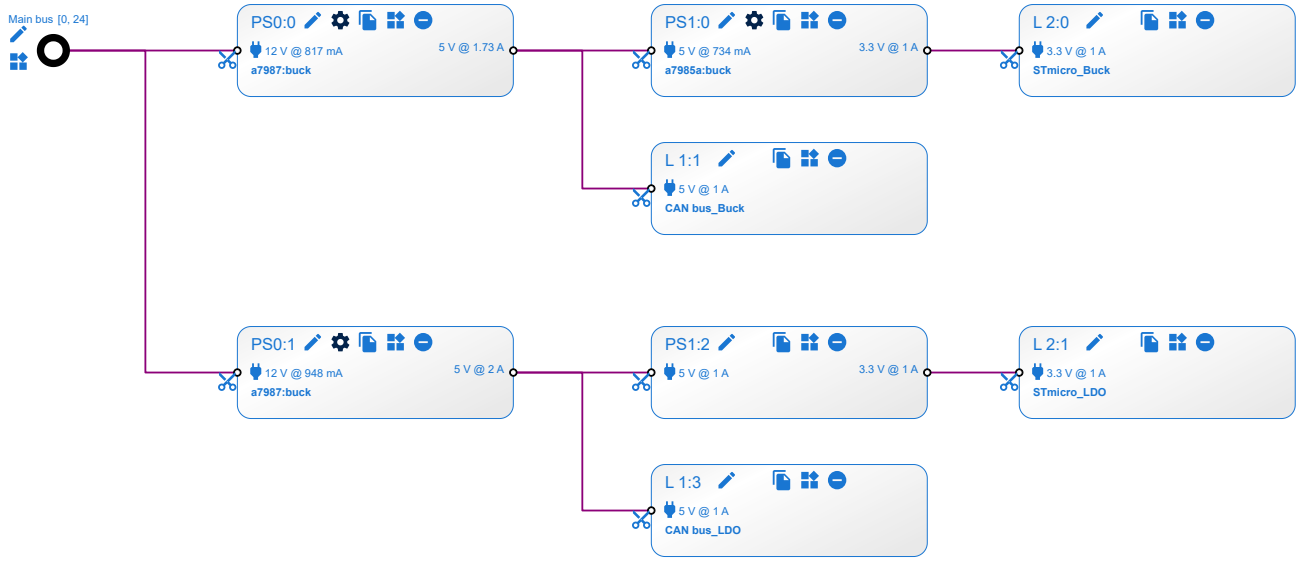
Project name : ESA Project

Power Tree: global schematic

Main bus

Input:: 12 V [0 - 24 V]
Vin:: 12 V
Pin:: 21.18 W
Iin:: 1.77 A
Pout:: 16.6 W
I/O Efficiency:: 78.37 %
Losses:: 4.58 W

Global Power Tree Schematic



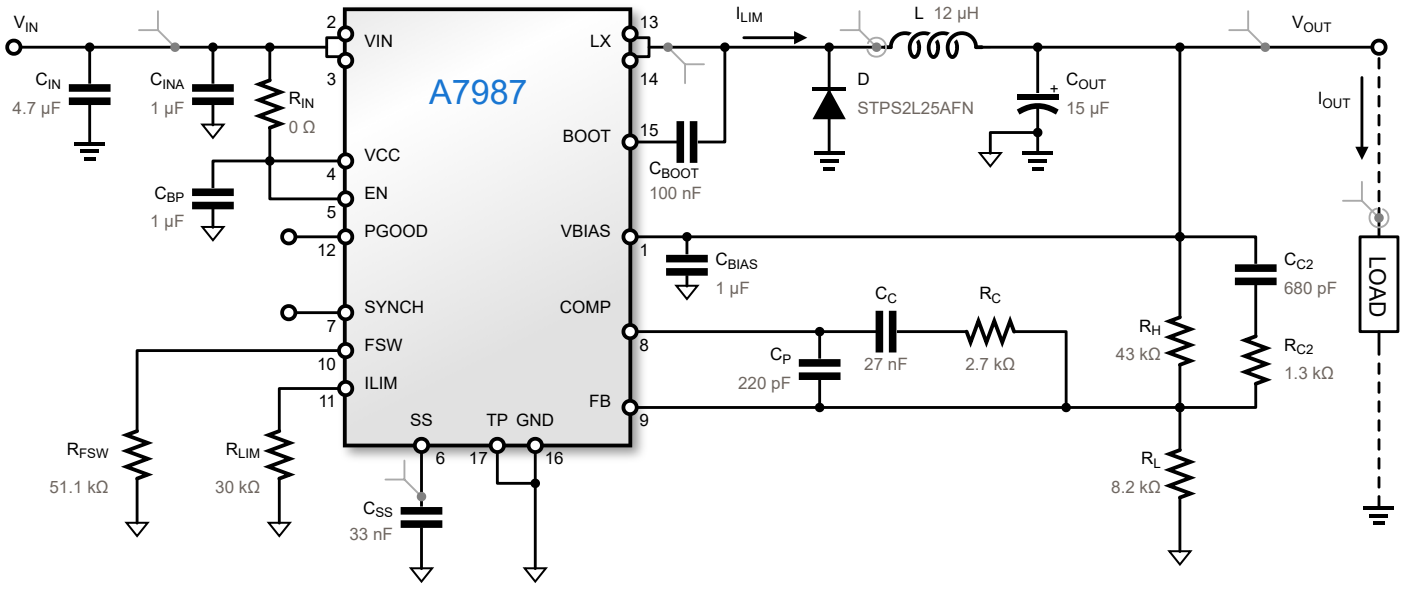
Results table by power branch

Stage 's name	*Vin*	*Iin*	*Pin*	*Total Efficiency*
can bus_buck	*5 V*	*1 A*	*5 W*	88.44 %
stmicro_buck	*3.3 V*	*1 A*	*3.3 W*	79.54 %
can bus_ldo	*5 V*	*1 A*	*5 W*	87.87 %
stmicro_ldo	*3.3 V*	*1 A*	*3.3 W*	58 %
Total			*16.6 W*	

Power Stage 0:0

Component type:: eDS Stage
IC:: A7987
Topology:: BUCK
Input:: 6 - 12 V
Output: 0: : 5 V (2 % ripple) - 1.73 A
Efficiency:: 88.44 % at [Vin: 12 V; Iout: 1.73 A; Ta: 25 °C]
Total Efficiency:: 88.44 %

Schematic

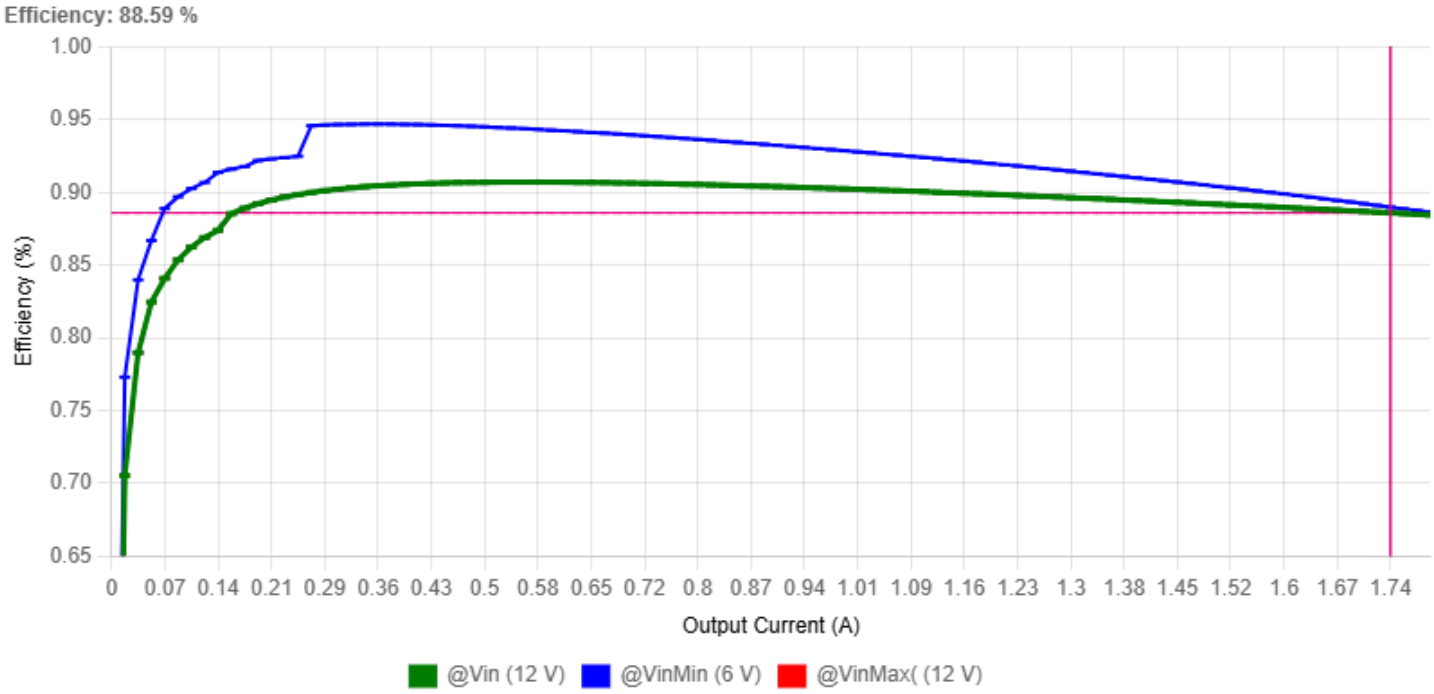


BOM

Type	Ref	Value	Description
IC	IC	A7987	A7987 - HTSSOP16 - STMicroelectronics
Capacitor	Cin	4.7 µF	16 V - 20% - Taiyo Yuden - EMK316BJ475ML
Capacitor	Cout	15 µF	6.3 V - 20% - Sanyo - 6SL15M
Inductor	L	12 µH	3.15 A - Würth Elektronik - 744776112
Diode	D	STPS2L25AFN	2 A, 25 V - STMicroelectronics
Capacitor	Cc	27 nF	27 nF
Resistor	Rc	2.7 kΩ	2.7 kΩ
Capacitor	Cp	220 pF	220 pF
Capacitor	Cc2	680 pF	680 pF
Resistor	Rc2	1.3 kΩ	1.3 kΩ

Type	Ref	Value	Description
Resistor	Rh	43 kΩ	Resistor value: 43 kΩ - tolerance: 1 %
Resistor	Rl	8.2 kΩ	Resistor value: 8.2 kΩ - tolerance: 1 %
Resistor	Rfsw	51.1 kΩ	51.1 kΩ
Capacitor	Cbp	1 µF	1 µF
Capacitor	Cbias	1 µF	1 µF
Capacitor	Css	33 nF	33 nF
Resistor	Rlim	30 kΩ	30 kΩ
Capacitor	Cboot	100 nF	100 nF
Capacitor	CinA	1 µF	1 µF
Resistor	Rin	0 Ω	0 Ω

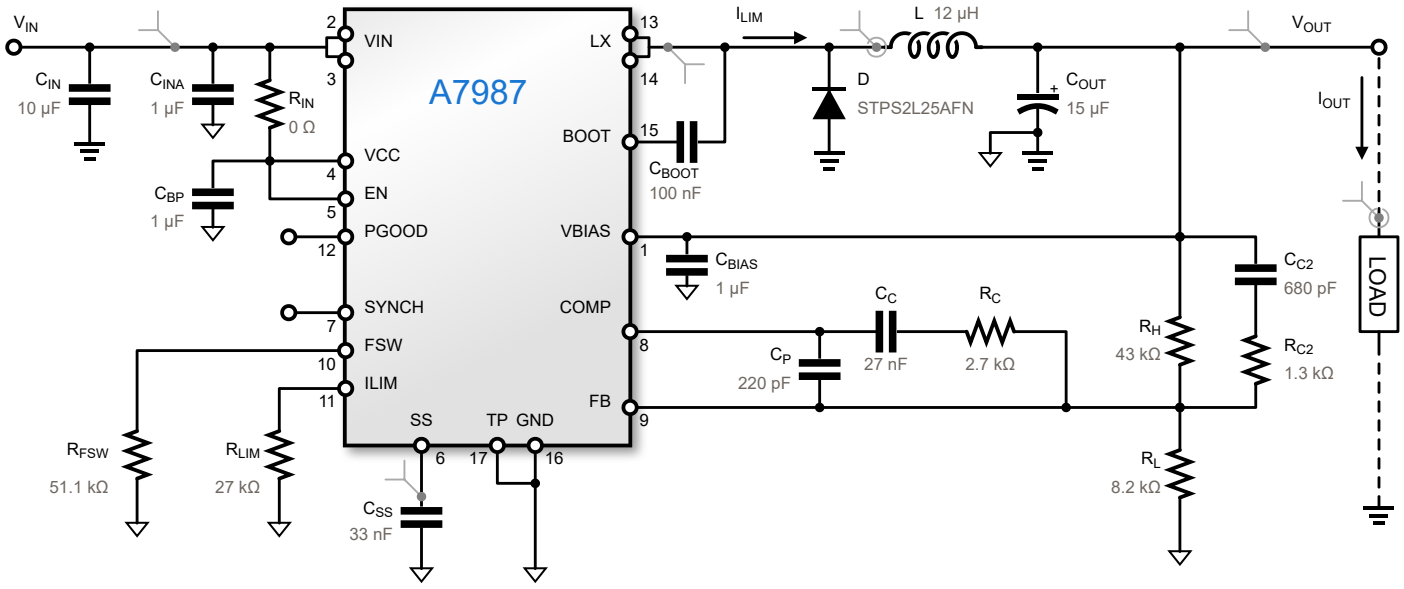
Efficiency chart



Power Stage 0:1

Component type:: eDS Stage
IC:: A7987
Topology:: BUCK
Input:: 6 - 12 V
Output: 0: : 5 V (2 % ripple) - 2 A
Efficiency:: 87.87 % at [Vin: 12 V; Iout: 2 A; Ta: 25 °C]
Total Efficiency:: 87.87 %

Schematic

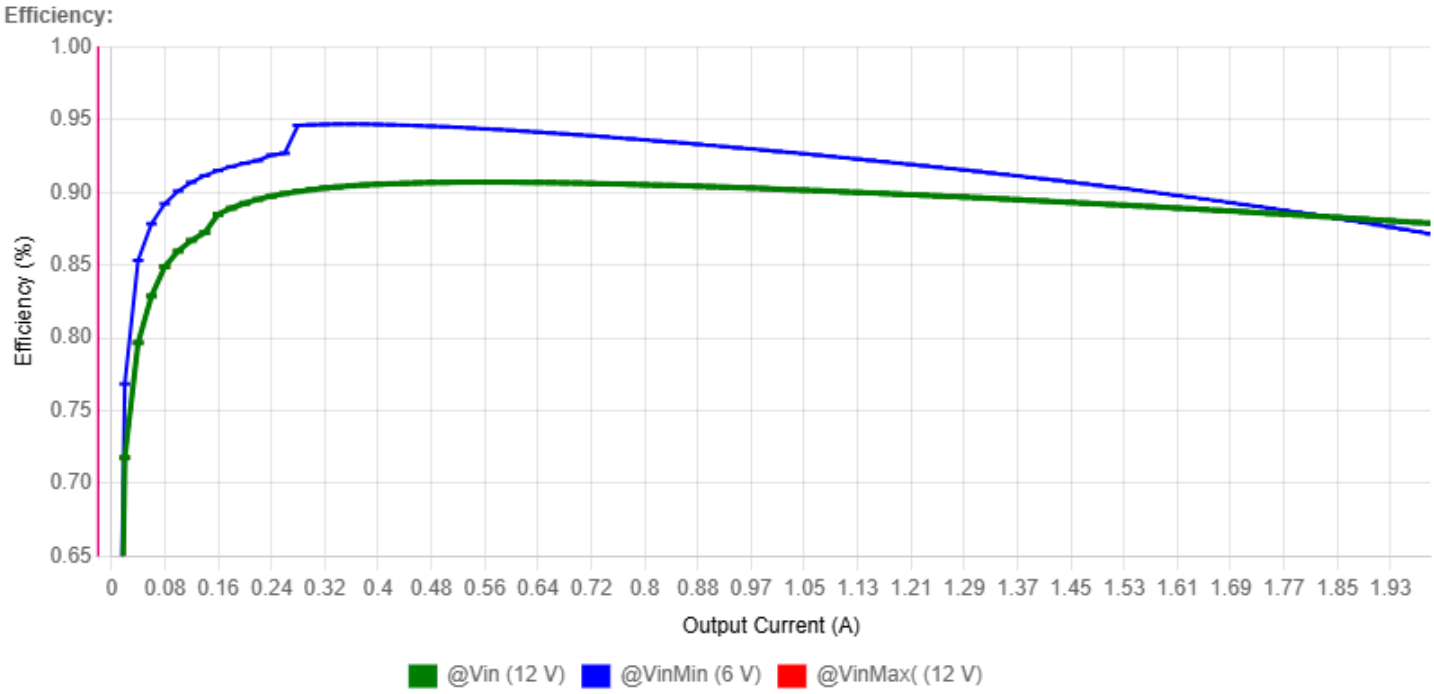


BOM

Type	Ref	Value	Description
IC	IC	A7987	A7987 - HTSSOP16 - STMicroelectronics
Capacitor	Cin	10 µF	16 V - 20% - Taiyo Yuden - EMK325BJ106KN-T
Capacitor	Cout	15 µF	6.3 V - 20% - Sanyo - 6SL15M
Inductor	L	12 µH	3.15 A - Würth Elektronik - 744776112
Diode	D	STPS2L25AFN	2 A, 25 V - STMicroelectronics
Capacitor	Cc	27 nF	27 nF
Resistor	Rc	2.7 kΩ	2.7 kΩ
Capacitor	Cp	220 pF	220 pF
Capacitor	Cc2	680 pF	680 pF
Resistor	Rc2	1.3 kΩ	1.3 kΩ

Type	Ref	Value	Description
Resistor	Rh	43 kΩ	Resistor value: 43 kΩ - tolerance: 1 %
Resistor	Rl	8.2 kΩ	Resistor value: 8.2 kΩ - tolerance: 1 %
Resistor	Rfsw	51.1 kΩ	51.1 kΩ
Capacitor	Cbp	1 µF	1 µF
Capacitor	Cbias	1 µF	1 µF
Capacitor	Css	33 nF	33 nF
Resistor	Rlim	27 kΩ	27 kΩ
Capacitor	Cboot	100 nF	100 nF
Capacitor	CinA	1 µF	1 µF
Resistor	Rin	0 Ω	0 Ω

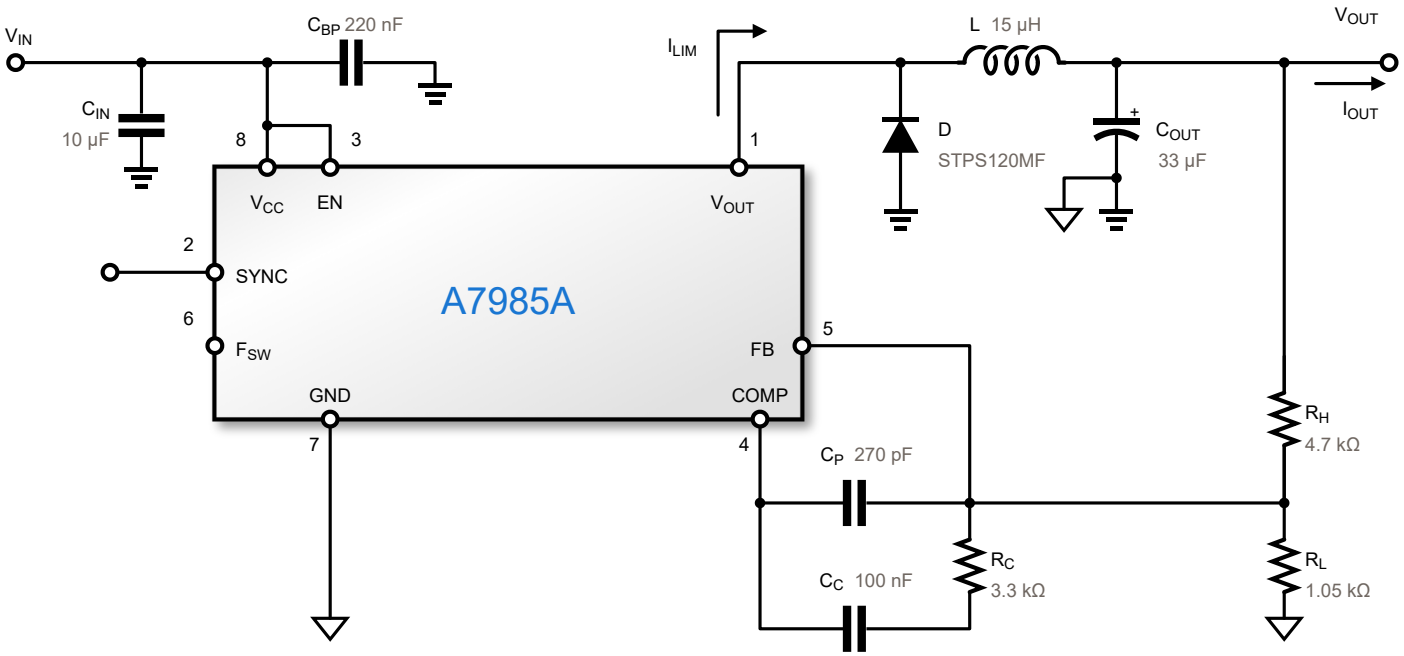
Efficiency chart



Power Stage 1:0

Component type:: eDS Stage
IC:: A7985A
Topology:: BUCK
Input:: 5 - 5 V
Output: 0: : 3.3 V (2 % ripple) - 1 A
Efficiency:: 89.94 % at [Vin: 5 V; Iout: 1 A; Ta: 25 °C]
Total Efficiency:: 79.54 %

Schematic

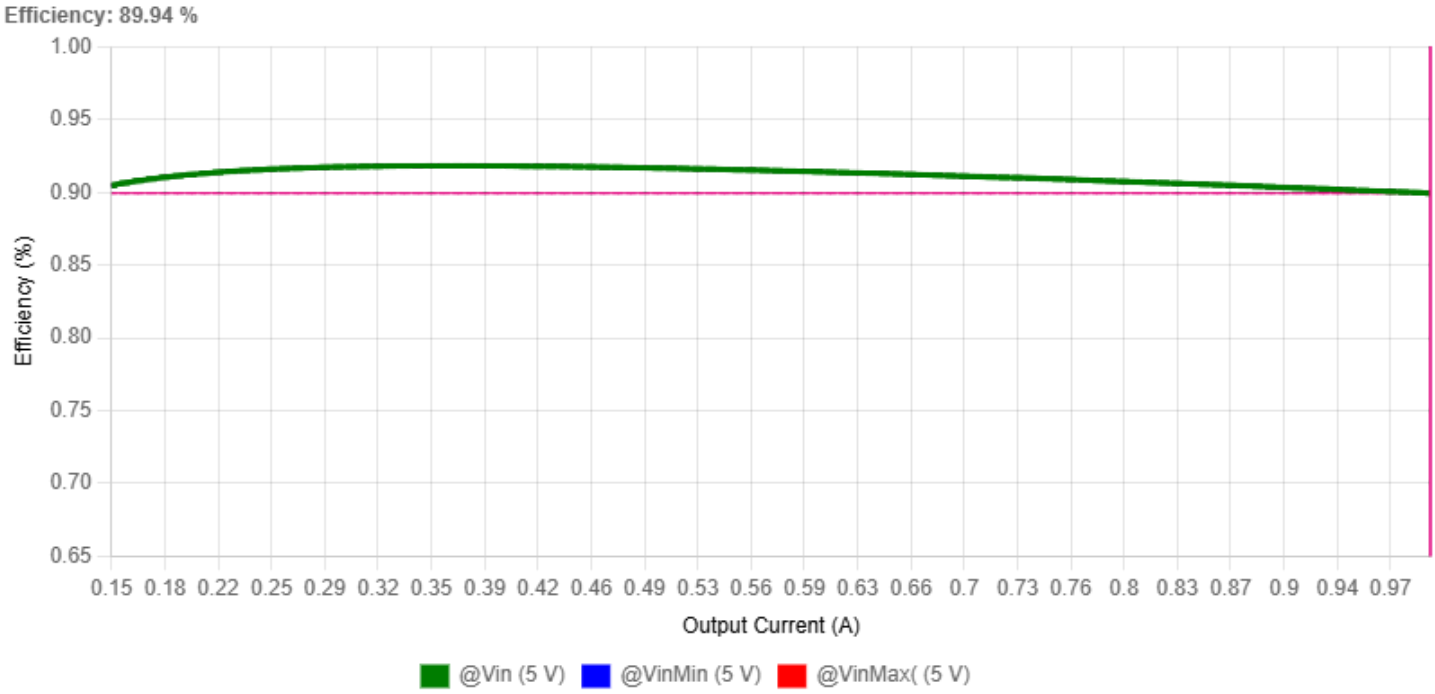


BOM

Type	Ref	Value	Description
IC	IC	A7985A	A7985A - HSOP 8 - STMicroelectronics
Capacitor	Cin	10 µF	6.3 V - 10% - muRata - GRM21BR60J106KE19L
Capacitor	Cout	33 µF	4 V - 20% - Panasonic - 4SVP33M
Inductor	L	15 µH	3 A - Würth Elektronik - 7447720150
Diode	D	STPS120MF	1 A, 20 V - STMicroelectronics
Capacitor	Cc	100 nF	100 nF
Resistor	Rc	3.3 kΩ	3.3 kΩ
Capacitor	Cp	270 pF	270 pF
Resistor	Rh	4.7 kΩ	Resistor value: 4.7 kΩ - tolerance: 1 %
Resistor	Rl	1.05 kΩ	Resistor value: 1.05 kΩ - tolerance: 1 %

Type	Ref	Value	Description
Capacitor	Cbp	220 nF	220 nF

Efficiency chart



Power Stage 1:1

Component type:: LDO Stage
IC:: STAGE_NAME_TIP
Description:: DESCRIPTION_TIP
Input:: 5 - 5 V
Output: 0: : 3.3 V - 1 A
Efficiency:: 66 %
Total Efficiency:: 58 %